

Henry Boyes

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Data & ML engineer with production experience building end-to-end pipelines — HPC orchestration, ETL, deep learning inference, and deployed data tools at clinical scale.

M.S. Data Science in progress (Pitt, 2027). **Navy QM background** applied to reproducible, maintainable systems.

TECHNICAL SKILLS

Languages & ML: Python (scikit-learn, statsmodels, pandas, NumPy), R, SQL

ML & Modeling: Supervised/unsupervised learning, deep learning inference, clustering (K-means, hierarchical, Leiden/UMAP), PCA, spatial statistics, feature engineering

Pipeline & Dev: Git, GitHub Actions, Slurm (HPC), ETL workflows, GPU compute (A100), SQLite, SQLAlchemy, CustomTkinter, openpyxl, BeautifulSoup

Visualization: matplotlib, seaborn, Tableau, Streamlit, Plotly

Process: Lean Six Sigma Yellow Belt, workflow optimization

PROFESSIONAL EXPERIENCE

Research Technician II – Cleveland Clinic Foundation

Mar 2025 – Present

- Designed and deployed a **production spatial proteomics ML pipeline** processing ~3M cells — GPU segmentation (Cellpose v4), Leiden clustering/UMAP, spatial analysis (Squidpy). github.com/boyeshenry-byte/akoya-pcf-pipeline
- Engineered **Slurm HPC orchestration** across GPU (A100) and CPU nodes — fully reproducible, minimal manual intervention.
- Built **Python ETL workflows cutting data prep 40%**; SQLite job tracking and centralized logging for pipeline observability and failure diagnosis.
- Replaced 47-sheet Excel log with **full-stack desktop app** (Python/CustomTkinter/SQLite) — multi-user, CRUD, filtered export, soft delete. github.com/boyeshenry-byte/lab-accession-app
- Applied **logistic & ordinal regression with bootstrapped CIs** (scikit-learn/statsmodels) to n=25 bladder cancer dataset; identified directionally significant ICI biomarkers.
- Managed **800+ oncology patient records**; enforced data quality, integrity, and HIPAA compliance across cross-functional teams.

Research Assistant – Kent State University

Aug 2023 – Oct 2024

- Contributed statistical analysis to a **peer-reviewed publication** in Acta Biomaterialia (Aug 2025); standardized analytical practices across the team.
- Presented complex findings at conferences and to cross-disciplinary audiences.

HM2, Quality Management – U.S. Navy, NMCS D

Dec 2018 – Apr 2020

- Led data modernization — **HEDIS metrics +60%** and training readiness **60%→95% in one month** via data-driven process redesign.
- Supervised 2 staff (both promoted); awarded **Navy Achievement Medal** and **Dept. Sailor of the Year (QM)**.

HM3 / BM3 – U.S. Navy, NMCS D & USS Porter (DDG 78)

Dec 2012 – Dec 2018

- Floor corpsman in Med/Surg ICU; maintained patient records and trained staff on documentation and compliance.
- Awarded **Arch Griffith Leadership Award**, **Blue Jacket of the Year**, and **Blue Jacket of the Quarter** for demonstrated leadership and operational excellence.

PROJECTS

Job Market Analyzer | Python, BeautifulSoup, SQLite, SQLAlchemy, GitHub Actions, Streamlit, Plotly | github.com/boyeshenry-byte/job_market_analyzer

- Automated multi-source scraping pipeline (3 platforms) with **GitHub Actions CI/CD scheduling** and deployed Streamlit dashboard for job market trend visualization.

EDUCATION & CERTIFICATIONS

University of Pittsburgh (Expected 2027) — M.S. Data Science | Coursework: Machine Learning, Statistical Analysis, Database Systems

Kent State University (2024) — B.S. Organismal Biology, cum laude (3.6 GPA) | A.S. Biology (3.4 GPA) |

Departmental Award for Significant Academic Achievement

Lean Six Sigma Yellow Belt (2019) | Project Management Performance Excellence (2019) | IBM AI Fundamentals (2025) | IBM Data Fundamentals (2025)

Previously held U.S. Secret Security Clearance